

FIN ST822–ST911

Recommended Paths Exhaustion
and the Local-Research Stop Theorem

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project

ORCID: 0009-0002-0909-3613

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All new-physics constructions are permitted when explicitly typed; none is silently promoted to strict provenance. No external evidence is introduced.

Abstract

ST822–ST911 execute the research paths recommended after the current-strict declared-class no-go. The cycle audits hard mathematical closures, a nonlinear nadsoliton equation, a seven-charge object, dynamic legacy-to-strict memory, metric-response gravity, gauge provenance and operational transfer.

No new strict object passes the seven-field intake gate requiring type, transformation law, source formula, nonzero witness, coupling, target independence and held-out transfer. A finite DNLS equation is mathematically coherent but leaves nonlinearity, pump, loss, seed, selector and refinement scaling unsourced. A Stieltjes–Schur memory bridge has the correct positivity/memory structure but no target-independent signed measure or cross-carrier prediction. Metric-response gravity can be constructed only after importing a metric-dependent weight law; conservation, anisotropic source, Newton/action unit, equivalence principle and held-out geometry remain absent. Gauge complexes remain conditional.

The mathematical frontiers of the first transition, degree-eight characteristic-zero relations, correlated IR intervals and full continuum bounds remain open but method-gated rather than conceptually confused. Operational software, validators and synthetic fixtures are ready; independent anchors, custody roles, apparatus and nature records are absent.

The conclusion is a local-research stop for repeated source scans. Further meaningful progress requires one of three state changes: a genuinely new formula-level strict object, a genuinely new exact mathematical method, or an independent external record. The conditional effective theory remains coherent and falsifiable, but strict physical/ToE closure remains absent.

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1 Intake rule

Every strict candidate was required to provide: type, transformation law, source formula, nonzero witness, coupling theorem, target independence and held-out carrier transfer. Zero candidates passed all seven rows.

2 Round I: ST822–ST836

The transition, algebra and IR lanes require respectively a new interval/SOS method, a structure-aware characteristic-zero method and correlated interval automatic differentiation. The continuum lane is conditionally ready for proof upgrades. No seven-charge object was supplied. Current A8 gravity and gauge provenance fail intake.

3 Round II: ST837–ST851

The conditional DNLS model

$$i\dot{\psi} = A\psi + g|\psi|^2\psi$$

has conserved norm/Hamiltonian and a uniform stationary branch. Defocusing energy is bounded below; focusing energy is controlled at fixed norm. Pump/loss can stabilize total norm conditionally. Yet g , pump, loss, seed, branch selector and continuum scaling are not sourced. A localized orbitally stable soliton was not proved.

4 Round III: ST852–ST866

The dynamic candidate

$$\Sigma(z) = A_{EH}(zI + A_{HH})^{-1}A_{HE}$$

is a completely monotone Stieltjes memory object and has the Schur static limit. Positive measures cannot reproduce the required signed oscillatory profile without signed resources. No measure passes provenance, target-independence and cross-carrier transfer; legacy physical roles remain nontransferable.

5 Round IV: ST867–ST881

A conditional law $g \mapsto A(g)$ permits formal Dirichlet metric variation and a stress candidate, but current artifacts supply no conserved nonproxy $T_{\mu\nu}$, anisotropic Bianchi-I source, Newton/action unit, equivalence-principle theorem or held-out geometry. The gravity acceptance matrix has zero complete rows. A8 remains partial/frozen.

6 Round V: ST882–ST896

Fine-rate tomography, two-clock, polarization/Gauss, causal-tail and signed-instability synthetic fixtures pass. Blind extraction, nuisance cover, one-shot unblinding and failure-stop rules are defined. Independent anchor, staffed custody roles, apparatus and laboratory events are absent.

7 Round VI: ST897–ST911

Frontier	Final status
First transition	open; needs new interval/SOS proof or counterexample
Degree-eight algebra	open; needs structure-aware exact $\mathbb{Q}$ method

Frontier	Final status
IR	open; needs correlated interval AD
Continuum cone	conditional mathematical proof upgrades remain possible
Nonlinear nadsoliton	conditional model; source/localization/stability open
Legacy–strict bridge	incomplete; memory type valid, instance absent
Gravity	frozen without new metric-response/source object
Gauge/Maxwell	conditional; scalar provenance absent
Operational transfer	software ready; apparatus/custody/data absent
Strict physical/ToE closure	blocked

8 Local-research stop theorem

Within all currently exhausted coarse-invariant source classes, further receiver scans, priors, generic actions, bootstrap maps, extra primes, ordinary multistarts, IR radius inflation, FRW residuals or target fits cannot export the missing strict physics. Repeating them is methodological replay.

Meaningful continuation requires:

1. a new formula-level strict object passing all seven intake rows;
2. a new exact method for one open hard-math frontier; or
3. an external independent apparatus/record state change.

This is not an absolute impossibility theorem for a future richer FIN object.

9 Recommended future action

Pause generic source-search batches. Maintain only:

- bounded work on transition/algebra/IR/continuum with genuinely new methods;
- intake of one explicit nonlinear or seven-charge strict object;
- external two-clock/fine-tomography transfer and independent audit.

A Reproducibility

The authoritative packets run from `FIN_ST822_ST836_Results.json` through `FIN_ST897_ST911_Results.json`, with six scripts, summaries and the combined hash test covering all 90 programmes.

B Nonclaims

No laboratory result, external referee, physical units, causal-light/Maxwell/gravity source, legacy role transfer, Standard Model, L_{total} or ToE closure follows. All future PDFs remain English.